



CAUÃ DE OLIVEIRA HOFFMANN

ELECTRICAL ENGINEERING & MECHATRONICS

Dual Degree | ENSEEIHT-INP Toulouse & UFSC

CONTACT

Toulouse, France
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SKILLS

Simulation & CAD

Ansys Maxwell, Ansys HFSS, Workbench, AutoCAD

Programming & PLC

Python, MATLAB, Ladder (PLC)

Instrumentation

B&K Connect, LMS TestLab, SCADAS

LANGUAGES

- Portuguese
- French
- English

COURSES

- ESSS: Python for Engineers
- ESSS: Electric Machines in Ansys Maxwell (advanced)

INTERESTS

- Electric Machines
- Mechatronics & Control
- Electromagnetic Simulation
- Energy Efficiency



PROFILE

Electrical engineer in a dual-degree program in France, with industrial R&D experience alongside Nidec and Electrolux. I combine numerical modeling (Ansys Maxwell), scientific programming (Python, MATLAB) and laboratory experimental validation, with a focus on electric machines, control and electromagnetic simulation.



EDUCATION

Master 2 – Electrical Engineering

2026 – PRESENT

ENSEEIHT – INP Toulouse | Dual Degree (BRAFIPEC)

- Parcours Énergie (2026.2) → Parcours Mécatronique (2027.1)

Bachelor's – Electrical Engineering

2022 – 2028 (EXPECTED)

Federal University of Santa Catarina (UFSC)

- Overall GPA (IAA): 8.87 / 10



R&D EXPERIENCE

R&D Research Fellow – Nidec / UFSC

JAN. 2025 – AUG. 2026

Vibration and Acoustics Laboratory

- Optimization of a Scroll compressor driven by an unbalanced two-phase induction motor
- 2D/3D simulations in Ansys Maxwell integrated with Workbench
- PLC (Ladder) automation of the test bench and efficiency tests with GRUCAD

R&D Research Fellow – Nidec / UFSC

JAN. 2024 – DEC. 2024

Vibration and Acoustics Laboratory

- Noise and vibration of PMSM motors for washing machines
- Tests in a semi-anechoic chamber (LMS TestLab) and data analysis in Python/MATLAB
- Simulation of radial forces and torque in Ansys Maxwell

R&D Research Fellow – Electrolux / UFSC

AUG. 2023 – DEC. 2023

Vibration and Acoustics Laboratory

- Noise mitigation in refrigeration systems, with measurements in reverberation and semi-anechoic chambers (B&K Connect)
- Technical reports for Electrolux engineering



PROJECTS

Electrical Design of a Residential Building

Complete low-voltage system: main distribution board, circuit division, protection and grounding.

Tools: AutoCAD, MS Excel

Chorus Effect Pedal Design

Guitar audio system: delay line with PT2399, LFO with TL072, anti-latchup circuit, LM78L05 regulation and 3PDT True Bypass switching.

Tools: LTSpice, KiCad